

The Chiral Dirac Equation Challenges: Hermiticity, Neutrino Mass Formalism, and Extensions to Higher Spin Fields.

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Abstract

The Chiral Symmetric Dirac Equation (DECS), derived from the irreducible representations of the Poincaré group, introduces an essential chiral angle (α) and an associated pseudoscalar mass term. This work addresses critical theoretical challenges arising from this generalization of the Dirac equation. We begin by formally establishing the physical Hermiticity of the DECS Hamiltonian by demonstrating that the complex mass term $m e^{-i2\alpha\gamma^5}$ can be reduced to a purely scalar mass through a global chiral field rotation, which ensures real eigenvalues for the system's energy. Subsequently, we resolve the inherent contradiction of deriving Dirac mass for the left-chiral neutrino field (νL) in the absence of a right-chiral component (νR). We propose a coherent mass mechanism rooted in the $\Delta L = 2$ violation of the Majorana formalism, providing a physically consistent theoretical framework for neutrino mass generation within the DECS context. Finally, we explore the conceptual role of the chiral degree of freedom in relativistic wave equations for fields with higher spin (e.g., Proca and Rarita-Schwinger), thereby laying the groundwork for future extensions of the chiral formalism beyond spin 1/2. Our findings affirm the consistency of the Chiral Dirac Equation and contribute to a more general understanding of chiral structure in Quantum Field Theory.

Keywords: Chiral Symmetry, Dirac Equation, Majorana Mass, Neutrino Mass, Hermiticity, Poincaré Group, Clifford Algebra, Pseudoscalar Mass, Higher Spin Fields, Yukawa Coupling, Spin 1/2.

1 Introduction

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2 Methods

We adopt the formalism of relativistic quantum field theory on a flat, non-interacting Minkowski spacetime. The background geometry is defined by the orthonormal basis $\{\gamma^\mu\}$ satisfying the Clifford algebra $\mathcal{Cl}_{1,3}(\mathbb{C})$, given by the anticommutation relation

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu}, \quad (1)$$

where $\eta^{\mu\nu}$ is the Minkowski metric. Consequently, $(\gamma^0)^2 = 1$ and $(\gamma^k)^2 = -1$, where Greek indices $\mu = 0, \dots, 3$ denote spacetime coordinates and Latin indices $k = 1, \dots, 3$ denote spatial components.

The chiral structure is determined by the pseudoscalar $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$, where i is the imaginary unit. Since the spacetime dimension is even, γ^5 anticommutes with all vector generators γ^μ and squares to the identity. This permits the construction of the orthogonal chiral projectors

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5), \quad (2)$$

satisfying $P_L + P_R = 1$ and $P_L P_R = 0$. A spinor field ψ therefore decomposes as $\psi = P_L \psi + P_R \psi \equiv \psi_L + \psi_R$.

2.1 Hamiltonian Formulation of the Chiral Dirac Equation

Following the formulation of Watson and Musielak [? ? ?], we consider the Chiral Dirac Equation (CDE) for a spinor field ψ :

$$(i\gamma^\mu \partial_\mu - m e^{-i\alpha\gamma^5})\psi = 0. \quad (3)$$

The corresponding effective Lagrangian density is given by

$$\mathcal{L}_{\text{eff}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m e^{-i\alpha\gamma^5})\psi, \quad (4)$$

where $\bar{\psi} = \psi^\dagger \gamma^0$ is the Dirac adjoint. We derive the Hamiltonian density $\mathcal{H}$ via the Legendre transform, using the canonical momentum $\pi = \partial \mathcal{L}_{\text{eff}} / \partial \dot{\psi}$:

$$\mathcal{H} = \pi \dot{\psi} - \mathcal{L}_{\text{eff}}. \quad (5)$$

Explicit calculation yields:

$$\mathcal{H} = \psi^\dagger (-i\gamma^{0k} \partial_k + \gamma^0 m e^{-i\alpha\gamma^5})\psi. \quad (6)$$

3 Hermitiiiiiiiiiii

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4 Chiral Seesaw Mechanism

As an alternative approach to explain the smallness of neutrino masses, we revisit the mass generation mechanism proposed by Watson and Musielak within the framework of the Chiral Dirac Equation (CDE). In this formalism, the fermion mass arises from Yukawa couplings to both a scalar field ϕ_1 (Standard Model-like) and a pseudoscalar field ϕ_2 .

In the original proposal, restricted to pure left-handed fermionic fields, the mass terms are shown to vanish or require fine-tuned cancellations. However, if we assume the existence of a right-handed neutrino component ν_R (such that $\nu = \nu_L + \nu_R$), the Dirac mass term does not vanish. Instead, it acquires a complex phase determined by the chiral angle α , which is intrinsic to the CDE. To naturally suppress the neutrino mass without relying on fine-tuning, we invoke the Majorana mass term, leading to a Chiral Seesaw Mechanism.

4.1 The Chiral Dirac Mass Term

Following Watson and Musielak, the general Yukawa Lagrangian expanded about the vacuum expectation values (VEVs) v_1 and v_2 is given by:

$$\mathcal{L}_Y \approx -\frac{\lambda_1 v_1}{\sqrt{2}} \bar{\nu} \nu - \frac{\lambda_2 v_2}{\sqrt{2}} \bar{\nu} \gamma^5 \nu. \quad (7)$$

Defining the scalar mass parameter $M = \lambda_1 v_1 / \sqrt{2}$ and the pseudoscalar mass parameter $\tilde{M} = \lambda_2 v_2 / \sqrt{2}$, and decomposing the fields into their chiral components ($\nu_{L,R} = P_{L,R} \nu$), the Dirac mass term becomes:

$$-\mathcal{L}_{\text{Dirac}} = \bar{\nu}_L (M + \tilde{M}) \nu_R + \bar{\nu}_R (M - \tilde{M}) \nu_L. \quad (8)$$

In the CDE formalism, these parameters are related to the propagation mass m and the chiral angle α by $M = m \cos 2\alpha$ and $\tilde{M} = -im \sin 2\alpha$. Consequently, the effective Dirac mass coupling M_D connecting $\bar{\nu}_L$ and ν_R is identified as:

$$M_D \equiv M + \tilde{M} = m(\cos 2\alpha - i \sin 2\alpha) = m e^{-2i\alpha}. \quad (9)$$

This result is crucial: the Dirac mass in this model carries a physical phase derived from the underlying chiral symmetry breaking.

4.2 The Full Majorana-Dirac Lagrangian

To implement the seesaw mechanism, we introduce Majorana mass terms. Unlike Dirac terms which couple left- and right-handed fields ($\bar{\nu}_L \nu_R$), Majorana mass terms violate lepton number by two units and must couple fields of the same chirality but opposite charge conjugation properties. Specifically, we utilize the charge-conjugated right-handed field $\nu_R^c = C \bar{\nu}_R^T$, which behaves as a left-handed field under Lorentz transformations.

The most general Lorentz-invariant mass Lagrangian including both the Chiral Dirac term derived above and the Majorana terms for left- and right-handed neutrinos is:

$$-\mathcal{L}_{\text{Mass}} = \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{M_L} + \mathcal{L}_{M_R} + \text{h.c.} \quad (10)$$

Explicitly writing the couplings:

$$-\mathcal{L}_{\text{Mass}} = \bar{\nu}_L M_D \nu_R + \frac{1}{2} \bar{\nu}_L^c M_L \nu_L + \frac{1}{2} \bar{\nu}_R^c M_R \nu_R + \text{h.c.} \quad (11)$$

Here, M_L represents the Majorana mass for the active left-handed neutrinos (assumed to be negligible or zero in the minimal Type-I seesaw), and M_R is the heavy Majorana mass scale for the sterile neutrinos.

4.3 The Chiral Mass Matrix

To find the physical mass eigenstates, we rewrite the Lagrangian using the basis of left-handed fields. We define the interaction basis spinor Ψ and its charge conjugate Ψ^c as:

$$\Psi = \begin{pmatrix} \nu_L \\ \nu_R^c \end{pmatrix}, \quad \text{and} \quad \Psi^c = \begin{pmatrix} \nu_L^c \\ \nu_R \end{pmatrix}. \quad (12)$$

Consequently, the adjoint of the charge conjugate is the row vector $\bar{\Psi}^c = (\bar{\nu}_L^c, \bar{\nu}_R)$. Utilizing the Majorana identity $\bar{\nu}_L M_D \nu_R = \frac{1}{2} \bar{\nu}_L M_D \nu_R + \frac{1}{2} \bar{\nu}_R^c M_D^T \nu_L^c$ (up to total derivatives), and the conjugated spinors identity, the Lagrangian allows the explicit matrix expansion:

$$-\mathcal{L}_{\text{Mass}} = \frac{1}{2} \underbrace{\begin{pmatrix} \bar{\nu}_L^c & \bar{\nu}_R \end{pmatrix}}_{\bar{\Psi}^c} \begin{pmatrix} M_L & m e^{-2i\alpha} \\ m e^{-2i\alpha} & M_R \end{pmatrix} \underbrace{\begin{pmatrix} \nu_L \\ \nu_R^c \end{pmatrix}}_{\Psi} + \text{h.c.} \quad (13)$$

This can be written in the concise matrix form:

$$-\mathcal{L}_{\text{Mass}} = \frac{1}{2} \bar{\Psi}^c \mathcal{M} \Psi + \text{h.c.}, \quad (14)$$

where the mass matrix $\mathcal{M}$ incorporates the chiral modifications from the CDE:

$$\mathcal{M} = \begin{pmatrix} M_L & M_D \\ M_D^T & M_R \end{pmatrix} = \begin{pmatrix} M_L & m e^{-2i\alpha} \\ m e^{-2i\alpha} & M_R \end{pmatrix}. \quad (15)$$

4.4 The Neutrino Chiral Mass

To account for the smallness of neutrino masses, it is necessary to consider that M_R corresponds to the mass of a right-handed sterile neutrino. In this framework, M_R represents the interaction term, while the chiral mass mixes the scalar and pseudoscalar components. We adopt the following identifications:

1. M_R (**Right-handed Majorana Mass**): Generated by a fundamental coupling. This represents the heavy scale, where $M_R \gg m$.
2. M_L (**Left-handed Majorana Mass**): We assume $M_L \approx 0$.

3. M_D (**Chiral Dirac Mass**): The complete mass term from the CDE corresponds to the coupling $\bar{\nu}_L \nu_R$, which we identify explicitly as $m e^{-2i\alpha}$.

Based on these assumptions, the mass matrix is given by:

$$\mathcal{M} = \begin{pmatrix} 0 & m e^{-2i\alpha} \\ m e^{-2i\alpha} & M_R \end{pmatrix}. \quad (16)$$

Finding the eigenvalues of the matrix in the usual way yields two mass terms. By expanding the characteristic equation and applying the seesaw approximation ($m \ll M_R$), we obtain:

1. **Heavy Mass (m'_1): Sterile Dominant State**

$$m'_1 \approx M_R + \frac{(m e^{-2i\alpha})^2}{M_R} = M_R + \frac{m^2 e^{-4i\alpha}}{M_R} \quad (17)$$

This Majorana state is dominated by the scale M_R (sterile neutrino).

2. **Light Mass (m'_2): Active Dominant State**

$$m'_2 \approx \frac{(m e^{-2i\alpha})^2}{M_R} = \frac{m^2 e^{-4i\alpha}}{M_R} \quad (18)$$

This Majorana state corresponds to the active neutrino that is physically observed. The resulting mass term exhibits a complex phase factor $e^{-4i\alpha}$ dependent on the chiral angle α .

4.5 Mass Suppression and CP Violation

The application of the Chiral Seesaw Mechanism yields the following expression for the active neutrino mass:

$$m'_\nu \approx \frac{m^2}{M_R} e^{-4i\alpha}. \quad (19)$$

This equation implies two significant physical results regarding the magnitude and the phase of the neutrino mass.

4.5.1 Mass Suppression

The magnitude of the active neutrino mass is given by:

$$|m'_\nu| \approx \frac{m^2}{M_R}. \quad (20)$$

This confirms the standard seesaw suppression mechanism: the observed mass is inversely proportional to the scale of New Physics, M_R . For instance, if we assume the electroweak scale $m \sim 100$ GeV and the observed neutrino mass scale $|m'_\nu| \sim 0.1$ eV, the mechanism necessitates a heavy Majorana scale of $M_R \sim 10^{14}$ GeV.

4.5.2 The Chiral Phase and CP Violation

A novel feature of this model is that the **phase** of the light neutrino mass is directly determined by the chiral angle α :

$$\text{Phase}(m'_\nu) = e^{-4i\alpha}. \quad (21)$$

This dependence has profound implications:

- **Leptonic CP Violation:** In a three-generation scenario, this complex phase translates into Majorana phases and potentially the Dirac phase within the PMNS mixing matrix. A value of $\alpha \neq k\pi/4$ (where k is an integer) implies a neutrino mass with a non-trivial complex phase, thereby introducing a novel and fundamental source of **leptonic CP violation**.
- **BSM Link:** This establishes a direct link between the theoretical angle α of the CDE and a physical observable (CP violation) mediated by the seesaw scale M_R . It suggests that the choice of the chiral basis becomes a physically observable property when lepton number violation occurs.

This constitutes a robust and novel result that successfully integrates the theoretical framework of the CDE with the phenomenological success of the Seesaw model.

5 Higger spin

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6 Results

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7 Conclusions

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